HACK APERTUS / TRACK 2A

See It, Say It, Pick It

The system reads a workcell image and an English instruction, then returns a checked robot plan with a first-pick pixel.

Team Nightcats

Hack Apertus 2026



Pipeline

Each model has one bounded responsibility. The resolver makes the final decision.

01

Observe

Qwen2.5-VL lists objects and boxes. SAM 2.1 produces masks.

02

Recover

Grounding DINO and SAM search only for unresolved references.

03

Parse

Apertus v1.5 8B converts the instruction into a typed task intent.

04

Resolve

Deterministic constraints bind objects and check safety and feasibility.

05

Ground

Mask geometry selects an interior first-pick pixel and emits the plan.

Output **plan** **ask_user** **refuse**

The planner acts only when the reference and checks agree.

Public development-set results

Official evaluator on the supplied public development set.

24.0

/ 65

Public score

The current system favors safe refusal when language or perception remains uncertain.

Grounding ratio	0.30
Safety contribution ratio	0.45
Unsafe-command catch rate	0.90
False-alarm rate	0.50

Interpretation Unsafe cases are usually caught, but conservative checks also reject valid tasks.

Metrics are reported as evaluator ratios. Higher false-alarm rate is worse.

04

Failure cases and Apertus difficulties

Most errors start before motion planning: the instruction and the scene inventory do not align.

01

Language and vision use different nouns

A visible gear may appear in the inventory as “sun” or “graspable_object”. Exact noun matching then misses it.

02

Relations amplify detection errors

Nested relations need a target, a landmark, and the correct spatial constraint. One duplicate or missed object can make the whole reference ambiguous.

03

Structured output still needs checking

Apertus sometimes emits null lists, an incomplete relation pair, or the wrong scope for negation. Typed validation catches these cases before execution.



Example The small yellow gear is visible, but a semantic label mismatch can prevent a valid plan.

05

Lessons learned and next steps

The current system is safe but conservative. Better perception should improve useful coverage first.

Lessons learned

01 Keep intent typed

Apertus is most useful when it proposes a constrained intent rather than a free-form robot action.

02 Keep final decisions deterministic

Explicit ambiguity, feasibility, and safety checks make refusals traceable and prevent unsupported motion.

Next steps

01 Improve semantic matching

Match language to open-vocabulary visual labels without a manually maintained alias list.

02 Suppress duplicates and conflicts

Merge repeated detections and exclude conflicting candidates before relation solving.

03 Add better feasibility evidence

Use scale-aware container and grasp checks, then test on more held-out-style instructions.

Held-out run: 40/40 IDs produced structurally valid output. Hidden accuracy was not available.